HECTOR GENARO PACHECO

Software Engineer | Full-Stack • Cloud • AI • IoT • Embedded Systems

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PROFESSIONAL SUMMARY

Software Engineer with a physical-world engineering background spanning full-stack web development, AWS cloud infrastructure, AI platforms, IoT edge computing, and industrial instrumentation & controls. Specialized in building auditable, scalable software applications—ranging from containerized AWS ECS Fargate microservices to real-time Raspberry Pi telemetry pipelines.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, SQL, C++

Frontend: React, Next.js, HTML5, CSS3, Modern CSS (Flex/Grid)

Backend & APIs: Node.js, Express, REST APIs, WebSockets, Zod

Cloud & Infra: AWS (ECS, ECR, RDS, Secrets Mgr, KMS, VPC), Docker, Vercel, Linux, Git

Data & AI: PostgreSQL, Prisma, Palantir Foundry, Palantir AIP, Agentic AI Workflows

Hardware & Industrial: Raspberry Pi 5, GPIO, GPS, Sensors, 4–20 mA Loops, PLC I/O

ENGINEERING PROJECTS

CalTrack — *Industrial Calibration SaaS & Cloud Platform* **React • Node • Express • PostgreSQL • AWS ECS**

- Architected and deployed a full-stack industrial calibration platform for tracking process plant instrument tags, parameter specifications, user authentication/RBAC, and persistent audit histories.
- Containerized Node.js API with Docker on AWS ECS Fargate, Amazon ECR, RDS PostgreSQL, Secrets Manager, KMS, and Vercel Edge.

FieldTrack AI — *Edge AI & Sensor Telemetry Platform* **Raspberry Pi 5 • Python • Node • React • WebSockets**

- Built an edge-computing telemetry platform combining Python hardware daemons, GPS serial (NMEA) parsing, PIR motion hardware interrupts, OpenCV vision detection, and a real-time React dashboard over WebSockets.

Bridge AI — *AI-Assisted Funding Intelligence Platform* **React • TypeScript • Express • Prisma • PostgreSQL**

- Engineered a funding intelligence platform using React, TypeScript, Express, Prisma ORM, and PostgreSQL with hash-verified data provenance and human-in-the-loop governance controls.

Pi Arcade OS — *Embedded Hardware & Software Gaming Platform* **Raspberry Pi • Python • Pygame • GPIO • I2C LCD**

- Built an embedded hardware/software gaming system in Python/Pygame featuring direct GPIO arcade microswitch wiring, software debouncing, 16x2 I2C LCD stats output, and passive buzzer sound synthesis.

TECHNICAL & INDUSTRIAL EXPERIENCE

Instrumentation & Controls Technician — *Process Control & Industrial Systems* **Field Engineering Experience**

- Calibrated, commissioned, and troubleshot 4–20 mA process transmitters (pressure, temperature, flow, level), PLC I/O wiring harnesses, terminal blocks, relays, and field instruments across 480VAC, 120VAC, and 24VDC power systems.
- Verified process control loop operation, signal integrity, and safety shutdowns while maintaining auditable calibration logs.

EDUCATION & PROFESSIONAL TRAINING

Western Governors University (WGU) — B.S. Software Engineering (In Progress, 2026–Present)

Palantir Technologies — Foundry & AIP Training (May 2026): Foundry, AIP workflows, agentic AI, data governance, business-process modeling, code repositories

Mirion Technologies — SU-890 iCAM Operations & Maintenance (Mar 2026, 16.00 Continued Education Credits)